MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549A1
AFT (M37 EXTRUDED) PROPELLANT GRAIN FOR

This amendment forms a part of Military Specification MIL-P-63203, dated 13 September 1983 and is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all departments and agencies of the Department of Defense.

PAGE 2

2.1.1: Standards, Military: Delete "MIL-STD-453" and substitute "MIL-STD-00453".

PAGE 4

3.5: Add the following new sub-paragraphs:

"3.5.3 Alternate Test Method.

3.5.3.1 Plus 140°F. The average of three (3) maximum compression readings from each lot shall be no less than 306 pounds and the average of three (3) deformation readings shall be no less than 48 percent when tested as specified in 4.5.5.1.

3.5.3.2 Minus 40°F. The average of three (3) maximum compression readings from each lot shall be no less than 8,427 pounds and the average of three (3) deformation readings shall be no less than 23 percent when tested as specified in 4.5.5.1."

3.8: Add new paragraph as follows:

3.8 Proving ground test (First Article Only). When tested in accordance with 4.5.4 all the range and safety requirements of MIL-P-50578 shall be met.

FSC 1320

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PAGE 5

4.3.1.1: Delete in its entirety and substitute the following:

"4.3.1.1 Process changes. The first article tests and inspections cited herein will qualify the particular processing procedures. Any major processing change made subsequent to the manufacture of the first article samples must be reviewed by the design agency who at their discretion may require part or all of the first article repeated."

4.3.2.1.2: Delete "for information only".

4.3.2.2: Delete in its entirety.

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4.3.2: Table I, Under Ballistic test: Change "3.2" to "3.8".

PAGE 9

4.4.1: Change "or more than one month scheduled production ..." to "or more than two months scheduled production ...".

PAGE 13

4.4.2.4:

Add the following Critical defect: "1 Radiographic inspection (A) 100%, 3.3.1, 4.5.1.1".

Delete: Major 103 in its entirety.

Add the following: "Note A. May be done prior to spiral wrap and OD machining."

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4.4.2.6:

Add the following Critical defect: "1 Excessive cracks, foreign material, or cavitation 100% 3.3.2 Visual".

Delete: Major 101 in its entirety.

Major 105: Add Note "(b)" to the paragraph reference. Add the following to the notes section: "Note b. Grain segments from the same lot may be mixed to meet weight requirements provided the minimum web requirement is met."

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4.4.3.2.2: Delete in its entirety and substitute the following:

"4.4.3.2.2 Grain inspection - Critical Defect. One sample of a completed grain shall be selected from each sublot produced and radiographed. A sublot shall be defined as a maximum of eight unsawed billets consecutively extruded and loaded onto a single grain buggy. In the event that the sample grain fails to meet the requirements of 3.3.1, the entire sublot shall be x-rayed and any additional grains failing to meet the requirement of 3.3.1 shall be removed from the lot. This test shall be conducted in accordance with 4.5.1.2."

4.4.3.3: Delete 4.4.3.3.1 and 4.4.3.3.2 and substitute the following:

"4.4.3.3.1 Plus 140°F - Major Defect. Three (3) samples will be randomly selected from the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for maximum compression load and deformation as specified in 4.5.5 or 4.5.5.1. The lot shall be rejected if the average of the three specimens from any billet fails to meet the requirements of 3.5.1 or 3.5.3.1.

4.4.3.3.2 Minus 40°F - Major Defect. Three (3) samples will be randomly selected from the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for maximum compression load and deformation as specified in 4.5.5 or 4.5.5.1. The lot shall be rejected if the average of the three specimens from any billet fails to meet the requirements of 3.5.2 or 3.5.3.2."

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4.5.1.1:

First sentence: Change "parallel to" to "perpendicular to".

Change: "MIL-STD-453" to MIL-STD-00453".

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4.5.1.1.1:

Second sentence: Delete "photograph selected defects".

Add the following at the end of the paragraph: "If the imiage quality indicator can not be in the field of view at all times then it shall be used at the beginning and end of each shift as well as at a rate of once every eight billets. Should any of the simulated defects not be apparent during any of the verification checks, all the billets since the last acceptable test shall be rejected."

4.5.1.2: Change "MIL-STD-453" to "MIL-STD-00453".

PAGE 20

4.5.4: Delete in its entirety and substitute the following:

"4.5.4 Proving ground tests. The 50 grain assemblies manufactured during first article shall be inspected per Table I and then shipped to a specified government loading facility. The grains shall be loaded into M549 projectiles (Inert warhead/Live motor body) and tested at a government test facility in accordance with 4.5.4.1. Forward grains to be loaded with the test samples must be in accordance with MIL-P-63202 and may themselves be first article samples.

4.5.4.1 Ballistic tests. The 50 projectiles from 4.5.4 shall be tested in accordance with the M549 Projectile Loading, Assembling and Packing specification (MIL-P-50578). 10 rounds shall be fired at -65°F, 10 rounds at +145°F (M203 charge only) 30 rounds at +70°F. Failure of the propelling grains to perform according to the criteria of MIL-P-50578 shall be cause for rejection of the first article sample. All test data and raw magnetic radar tapes shall be sent to Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001."

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4.5.5: Add the following:

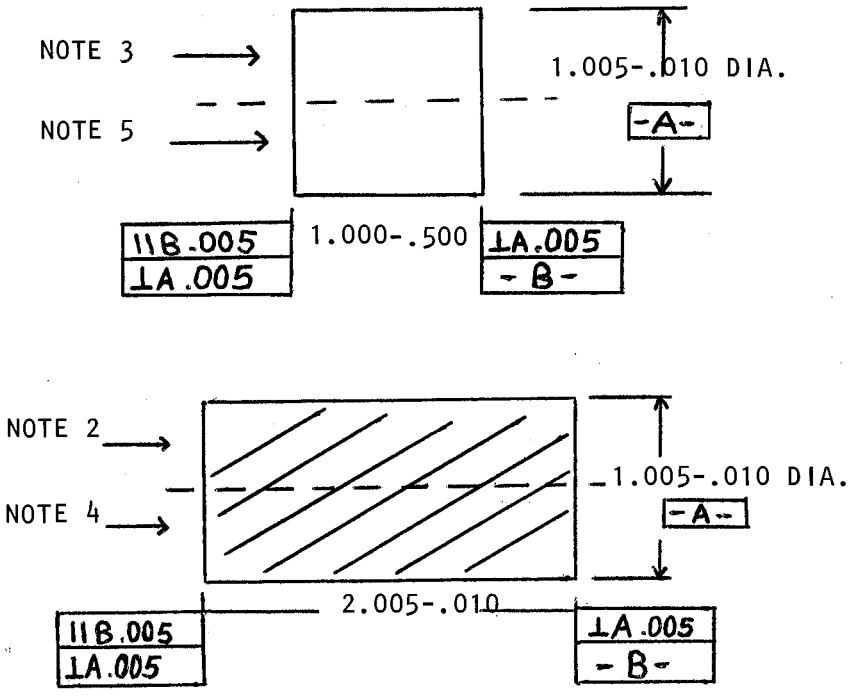
"4.5.5.1 Physical properties - alternate test method. Each specimen and specimen support shall conform to the requirements of Figure 1. The specimen shall be conditioned at the required temperature plus or minus 5°F for three hours, minimum. The specimen shall be removed from the conditioning box and inserted between the fixture posts with steel cylinders the same diameter as the specimen placed above and below the specimen forming one continuous concentric cylinder. The specimen shall be tested using a cross head speed of 2 inches per minute. The total time elapsed from removing the specimen from the conditioning box to the completion of each test shall be 1 ± 1/2 minute. All compression tests shall be conducted in a room that is maintained at 75°F ± 5°F."

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Army-AR

Preparing activity:
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(Project 1320-A830)

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NOTES:

1. SPECIFICATION MIL-A-2550 APPLIES.
2. SPECIMEN MATERIAL-SOLID PROPELLANT.
3. TWO STEEL SPECIMEN SUPPORT REQUIRED & SHALL BE PLACED ABOVE & BELOW SPECIMEN PRIOR TO TEST.
4. SURFACE FINISH-125/ALL OVER.
5. SURFACE FINISH-125 ON PROPELLANT SURFACE.

TITLE:
RAP GRAIN SPECIMEN & SPECIMEN SUPPORTS FOR COMPRESSION TEST

FIGURE 1

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